

A • PRODUCTION OF SOUND

Sound	A form of energy produced by vibrating objects — needs a MEDIUM to travel
Vibration	Rapid to and fro motion of a particle about its mean position — also called oscillation
Medium	Substance through which sound travels from the source to the listener
Production	Sound is produced when an object is SET TO VIBRATE — stops when vibration stops

- Plucked rubber band vibrates → humming sound | stops vibrating → sound stops
- Struck metal pan vibrates → sound produced; holding the pan stops vibration → sound stops
- Stringed instruments (guitar, sitar, veena): vibrating strings produce sound
- Percussion instruments (drum, tabla): striking membrane produces vibration and sound
- Sound waves travel outward from the vibrating source in ALL directions through the medium
- Thomas Alva Edison invented the phonograph in 1877 — first device to record and play back sound

B • PROPAGATION OF SOUND — Medium, Speed & Wave Nature

Sound needs medium	Sound CANNOT travel in vacuum — it needs a material medium (solid, liquid, gas)
Speed in solids	Maximum — e.g., Aluminium = 6420 m/s, Steel = 5960 m/s, Iron = 5950 m/s
Speed in liquids	Medium — Sea water = 1530 m/s, Distilled water = 1498 m/s
Speed in gases	Minimum — Air $\approx$ 330–344 m/s (at 0°C: 331 m/s; at 22°C: 344 m/s)
Speed order	Solid > Liquid > Gas (Sound travels fastest in solids)
Effect of temperature	Speed of sound INCREASES with increase in temperature
Effect of humidity	Speed of sound INCREASES with increase in humidity (density of air decreases)

Speed of Sound in Various Media at 25°C

State	Substance	Speed (m/s)
Solid	Aluminium	6420
Solid	Steel	5960
Solid	Iron	5950
Liquid	Sea Water	1530
Liquid	Distilled Water	1498
Gas	Hydrogen	1284
Gas	Air (at 25°C)	346
Gas	Oxygen	316

- Sound CANNOT travel in vacuum — bell jar experiment proves: as air is pumped out, sound fades
- On the Moon there is no atmosphere — two astronauts cannot talk to each other directly (radio waves used)
- Light travels much faster than sound — that is why lightning is seen before thunder is heard
- Astronauts: helmet devices convert sound waves to radio waves for communication
- Compression (C): region of HIGH pressure when tuning fork moves forward
- Rarefaction (R): region of LOW pressure when tuning fork moves backward
- Sound propagates as a series of compressions and rarefactions through the medium

C · SOUND WAVES — Wave Nature & Types of Mechanical Waves

Mechanical Wave	Disturbance that propagates through a medium due to repeated periodic motion of particles
Wave motion	Only ENERGY is transferred — NOT the particles of medium
Wavelength (λ)	Distance between two consecutive particles in same phase of vibration — unit: metre (m)
Frequency (n)	Number of vibrations of a particle in one second — unit: Hertz (Hz)
Amplitude (A)	Maximum displacement of a vibrating particle from its mean position — unit: metre (m)
Time Period (T)	Time taken by a vibrating particle to complete ONE vibration — unit: second (s)
Speed formula	$v = n \times \lambda$ (Speed = Frequency $\times$ Wavelength)

Speed of Sound

$$v = n \times \lambda \quad (v = \text{speed (m/s)}, n = \text{frequency (Hz)}, \lambda = \text{wavelength (m)})$$

Types of Mechanical Waves — Transverse vs Longitudinal

Type	Particle Movement	Direction vs Wave	Medium	Examples
Transverse Wave	Vibrates PERPENDICULAR to direction of wave propagation	Perpendicular (90°)	Solids and Liquids ONLY (not gases)	Light waves, waves in strings, surface water waves
Longitudinal Wave	Vibrates PARALLEL to direction of wave propagation	Parallel (same)	Solids, Liquids AND Gases	SOUND waves, waves in springs, seismic P-waves

■ Sound waves are **LONGITUDINAL** waves — particles vibrate parallel to wave direction. Sound cannot be transverse.

- Seismic wave: formed during earthquake — longitudinal wave travelling through Earth's layers
- Seismology: branch of science that studies seismic waves using seismometer
- Hydrophone: instrument to detect sound/seismic waves underwater
- In wave motion, the velocity of wave is **DIFFERENT** from velocity of vibrating particle
- Medium must have: inertia, elasticity, uniform density and minimum friction — for mechanical wave propagation

D · PROPERTIES OF SOUND — Loudness, Pitch, Quality, Audibility

Property	Definition	Depends On	Examples
Loudness	Characteristic that distinguishes a faint (weak) sound from a loud sound	AMPLITUDE — Higher amplitude → Louder sound Unit: decibel (dB)	Softly beaten drum = weak sound Strongly beaten drum = loud sound
Pitch	Characteristic that distinguishes a flat (low) sound from a shrill (high) sound	FREQUENCY — Higher frequency → Higher pitch (shriller sound)	Whistle, bell, flute, violin = high pitch Lion's roar, drum beat = low pitch
Quality (Timbre)	Characteristic that distinguishes two sounds of same pitch and amplitude	Waveform / overtones present in the sound	Sounds of guitar vs violin at same pitch still sound different — due to timbre

- Female voice has **HIGHER PITCH** than male voice — females have thinner, shorter vocal cords
- Male voice has **LOWER PITCH** — males have thicker, longer vocal cords → deeper sound

- Orchestra: different instruments produce sounds of same pitch but different timbre → identifiable
- Amplitude doubled → loudness does NOT double — relationship is logarithmic
- Amplitude quadrupled → loudness approximately doubles (in terms of decibels)

E · FREQUENCY RANGES — Audible, Infrasonic & Ultrasonic Sound

Type	Frequency Range	Heard By	Uses / Applications
Infrasonic (Subsonic)	Below 20 Hz	NOT audible to humans Dogs, dolphins, elephants, whales can sense	• Earth monitoring / seismic study • Study of heart mechanism • Elephants communicate over long distances
Sonic (Audible)	20 Hz to 20,000 Hz (20 kHz)	Human beings — the normal hearing range	• All normal communication and music • Audible range called "sonic range"
Ultrasonic (Supersonic)	Above 20,000 Hz (20 kHz)	NOT audible to humans Bats, dogs, dolphins can hear	• Medical: sonogram/ultrasound scan • SONAR (detect submarines, sea depth) • Galton's whistle (dog training) • Dish washers • ECHOLOCATION in bats

■ **Human Audible Range: 20 Hz to 20,000 Hz. Below 20 Hz = Infrasonic. Above 20,000 Hz = Ultrasonic.**

- Bat echolocation: bats produce ultrasonic sound (>20,000 Hz) → reflects from objects → bat detects prey and navigates in dark
- Galton's whistle: produces ultrasonic sound — inaudible to humans but heard by dogs — used for dog training
- Sonogram: ultrasound imaging of internal organs (foetus, liver, kidney) — non-invasive medical diagnosis
- Infrasound: produced naturally by earthquakes, volcanoes, ocean waves, heavy winds, tornadoes
- Dolphins use echolocation (ultrasound) to navigate and communicate underwater

F · ECHO, REVERBERATION & SONAR

Reflection of Sound	Sound bounces back when it hits a hard, large surface — follows same laws as reflection of light
Echo	Reflected sound heard SEPARATELY after original sound — minimum distance from wall = 17.2 m (approx. 17 m)
Minimum Echo Distance	Speed of sound = 344 m/s Ear needs 0.1 s to detect echo separately Distance = $344 \times 0.1 / 2 = 17.2$ m

Reverberation	Persistence of sound due to MULTIPLE reflections — sound continues after source stops; heard in closed halls
Reverb vs Echo	Echo: single distinct reflection heard separately Reverberation: multiple rapid reflections blending together

Applications of Reflection of Sound

Megaphone / Loudspeaker	Funnel-shaped; reflects sound waves and channels them forward — prevents spreading in all directions
Concert Halls / Auditoriums	Ceiling and walls curved to uniformly distribute reflected sound; absorbing materials reduce reverberation
Stethoscope	Uses multiple reflections inside tube to carry heart/lung sound to doctor's ear
Sound boards	Curved boards placed behind speakers in auditoria to reflect sound toward audience

SONAR — Sound Navigation And Ranging

SONAR	Sound Navigation And Ranging — uses ULTRASOUND to detect and locate objects underwater
Principle	Ultrasound pulses sent from ship → reflect from sea floor / submarine / fish → received back → time calculated
Formula	$d = v \times t / 2$ (d = depth, v = speed of sound in water, t = time for echo to return)
Uses	(1) Measure depth of sea (2) Detect submarines (3) Detect icebergs (4) Locate fish shoals (5) Survey sea floor
Speed in seawater	≈ 1530 m/s at 25°C

SONAR Depth Formula $d = (v \times t) / 2$ (d = depth (m), v = speed in water (m/s), t = total time for echo (s))

G · MUSICAL INSTRUMENTS — Four Types

■ Music = sound produced by REGULAR, PERIODIC vibrations | Noise = sound produced by IRREGULAR, NON-PERIODIC vibrations

Type	How Sound Produced	Frequency Control	Examples
Wind Instruments	Vibration of air column in a hollow tube	Changing the LENGTH of vibrating air column	Trumpet, Flute, Shehnai, Saxophone, Nadaswaram
Reed Instruments	Reed vibrates when air is blown through — reed vibration produces specific sound	Varying air pressure and reed position	Harmonium, Mouth Organ (Harmonica)
Stringed Instruments	Vibration of stretched string/wire — hollow box amplifies the sound produced	Varying length, tension and thickness of string	Violin, Guitar, Sitar, Veena, Harp
Percussion Instruments	Struck, scraped or clashed together — oldest type of musical instrument	Tension of stretched membrane / size of resonator	Drum, Tabla, Mridangam, Cymbals, Xylophone

- Guitar/Violin strings have natural frequencies called HARMONICS — depends on tension, linear density and length of string
- Percussion: drum/tablā has leather membrane stretched across hollow box called RESONATOR
- Harmonics: natural frequencies at which a string vibrates — multiple of fundamental frequency
- Keyboard (piano) = stringed instrument (hammers strike strings) — not percussion

H · HUMAN VOICE & HUMAN EAR — Structure & Function

Human Voice — Production of Sound by Humans

Voice Box (Larynx)	Organ of voice — located in throat at upper end of windpipe (trachea)
Vocal Cords	Two ligaments stretched across the larynx — vibrate when air passes through them
Narrow Slit	Gap between the vocal cords through which air from lungs is forced
How voice is produced	Air from lungs → pushed through trachea → passes through slit between vocal cords → vocal cords vibrate → sound
Pitch control	Varying THICKNESS of vocal cords changes the length of air column in slit → different pitches
Male voice lower	Males have THICKER and LONGER vocal cords → lower, deeper pitch than females
Female voice higher	Females have THINNER and SHORTER vocal cords → higher, shriller pitch

Human Ear — Structure & Function		
Part	Location	Function
Pinna	Outer ear — visible curved structure	Collects sound waves from environment and directs them into the ear canal
Ear Canal (Auditory Canal)	Connects pinna to eardrum	Conducts sound waves from pinna to the tympanic membrane
Eardrum (Tympanic Membrane)	Between outer and middle ear	Vibrates when sound waves strike it — converts sound waves to mechanical vibrations
Ossicles	Three tiny bones in middle ear: Malleus, Incus, Stapes	Amplify and transmit vibrations from eardrum to inner ear — move inward and outward
Cochlea	Inner ear — spiral/snail-shaped fluid-filled structure	Contains special sensory cells that convert mechanical vibrations to nerve signals
Eustachian Tube	Connects middle ear to throat	Equalises air pressure on both sides of eardrum
Auditory Nerve	Connects inner ear to brain	Transmits nerve signals from cochlea to brain for interpretation as sound
Semicircular Canals	Inner ear	Responsible for balance and equilibrium (not directly for hearing)

- Sound wave → Pinna → Ear Canal → Eardrum (vibrates) → Ossicles (amplify) → Cochlea (converts to signals) → Auditory Nerve → Brain
- Brain PERCEIVES these nerve signals as sound — hearing is a sensory process
- Aquatic animals' ears designed to pick up high-frequency vibrations in WATER
- Ear ache, fullness/fluid in ear, ringing in ears = symptoms of hearing loss
- Causes of hearing loss: aging, ear infections (untreated), certain medicines, genetic disorders, severe blow to head, loud noise

I · NOISE POLLUTION — Causes, Effects & Control	
Noise	Any UNPLEASANT, unwanted, irritating and loud sound — produced by IRREGULAR and NON-PERIODIC vibrations
Noise Pollution	Disturbance in the environment due to loud and harsh sounds from various sources

Major source

INDUSTRIES — noise pollution is a by-product of industrialisation, urbanisation and modern civilisation

Causes of Noise Pollution

- Busy roads and vehicle horns — excessive honking is a major urban noise source
- Airplanes and airports — jet engine noise
- Electrical appliances at home — mixer grinder, washing machine, un-tuned radio
- Loudspeakers during festivals, religious and political events
- Crackers/fireworks during festivals
- Industrial machines — factories, construction sites

Health Hazards of Noise Pollution

- Irritation, stress, nervousness and headache
- Changes sleeping pattern — long term exposure disturbs sleep
- Sustained exposure → affects hearing ability → may lead to permanent hearing loss
- Sudden loud noise → heart attack and unconsciousness
- Lack of concentration in work — horns, loudspeakers cause disturbances
- Peace of mind disturbed → high blood pressure, short-tempered nature

Measures to Control Noise Pollution

- Strict guidelines for use of loudspeakers on social, religious and political occasions
- All automobiles should have effective silencers
- People should refrain from excessive honking while driving
- Industrial machines and home appliances should be properly maintained
- All communication systems operated in low volumes
- Residential areas should be free from heavy vehicles
- Green corridor belt set up around industries (as per Pollution Control Board regulations)
- People working in noisy factories should wear ear plugs
- Planting trees and using absorbing materials like curtains and cushions at home

J • Solved Numerical Problems

Problem 1

A sound has a frequency of 50 Hz and a wavelength of 10 m. What is the speed of sound?

Given	$n = 50 \text{ Hz}; \lambda = 10 \text{ m}$
Solution	$v = n \times \lambda = 50 \times 10 = 500 \text{ m/s}$
Answer	500 m/s

Problem 2	A sound has a frequency of 5 Hz and a speed of 25 m/s. Find the wavelength.
Given	$n = 5 \text{ Hz}; v = 25 \text{ m/s}$
Solution	$v = n \times \lambda \rightarrow \lambda = v/n = 25/5 = 5 \text{ m}$
Answer	Wavelength = 5 m

Problem 3	A wave with a frequency of 500 Hz is traveling at 200 m/s. Find the wavelength.
Given	$n = 500 \text{ Hz}; v = 200 \text{ m/s}$
Solution	$\lambda = v/n = 200/500 = 0.4 \text{ m}$
Answer	Wavelength = 0.4 m

Problem 4	A sound wave travels 2000 m in 8 seconds. What is its velocity?
Given	$d = 2000 \text{ m}; t = 8 \text{ s}$
Solution	$v = d/t = 2000/8 = 250 \text{ m/s}$
Answer	Velocity = 250 m/s

Problem 5	Ruthvik and Ruha hear a gunshot 2 seconds after it is fired. How far are they from the gun? (Speed of sound in air = 330 m/s)
Given	$t = 2 \text{ s}; v = 330 \text{ m/s}$
Solution	$d = v \times t = 330 \times 2 = 660 \text{ m}$
Answer	Distance = 660 m

Problem 6	A SONAR sends ultrasound and receives the echo in 6 seconds. What is the depth of the sea? (Speed of sound in seawater = 1530 m/s)
Given	$v = 1530 \text{ m/s}; t = 6 \text{ s}$ (total time for echo)
Solution	$d = v \times t / 2 = 1530 \times 6 / 2 = 9180 / 2 = 4590 \text{ m}$
Answer	Depth of sea = 4590 m = 4.59 km

Formulas, Values & Key Numbers

Quantity / Concept	Formula / Value	Unit / Note
Speed of sound formula	$v = n \times \lambda$ (frequency $\times$ wavelength)	m/s
SONAR depth formula	$d = v \times t / 2$	metres
Speed in air at 0°C	331 m/s	Temperature dependent
Speed in air at 22°C	344 m/s	Increases with temp
Speed in aluminium	6420 m/s (fastest common medium)	Solid
Speed in steel	5960 m/s	Solid
Speed in sea water	1530 m/s	Liquid
Speed in distilled water	1498 m/s	Liquid
Speed in oxygen (gas)	316 m/s	Gas
Audible range (humans)	20 Hz to 20,000 Hz (20 kHz)	Sonic sound
Infrasonic	Below 20 Hz	Subsonic
Ultrasonic	Above 20,000 Hz	Supersonic
Minimum echo distance	≈ 17.2 m (from reflecting surface)	At 344 m/s, 0.1 s separation
Loudness unit	Decibel (dB)	Amplitude dependent
Wavelength unit	Metre (m)	λ (lambda)
Frequency unit	Hertz (Hz) = vibrations per second	n
Amplitude unit	Metre (m)	A
Time period unit	Second (s)	T

Scientists, Discoveries & Years

Scientist / Event	Contribution / Fact	Year
Thomas Alva Edison	Invented phonograph (first device to record and play back sound)	1877
Daniel Bernoulli	Studied harmonics of vibrating strings — natural frequencies	18th century

Scientist / Event	Contribution / Fact	Year
Francis Galton	Invented Galton's whistle — ultrasonic whistle for dog training	1876
SONAR invention	Used during World War I to detect submarines underwater	~1915-1918
Speed of sound (air)	First accurate measurement by Pierre Gassendi and Marin Mersenne	1635-1636
Eustachius	Eustachian tube named after Italian anatomist Bartolomeo Eustachi	16th century

L · Glossary — Definitions

Sound — A form of energy produced by vibrating objects; needs a material medium to travel	Vibration — Rapid to-and-fro (oscillatory) motion of a particle about its mean position
Medium — Substance through which sound energy is transmitted from source to listener	Compression (C) — Region of HIGH pressure in a sound wave — when vibrating body moves forward
Rarefaction (R) — Region of LOW pressure in a sound wave — when vibrating body moves backward	Mechanical Wave — Disturbance propagating through a medium due to periodic motion of particles
Transverse Wave — Wave in which particles vibrate perpendicular to direction of wave propagation; in solids and liquids only	Longitudinal Wave — Wave in which particles vibrate parallel to direction of propagation; sound is a longitudinal wave
Wavelength (λ) — Distance between two consecutive particles in same phase of vibration; unit: metre	Frequency (n) — Number of vibrations per second; unit: Hertz (Hz)
Amplitude (A) — Maximum displacement of vibrating particle from mean position; unit: metre	Time Period (T) — Time for one complete vibration; $T = 1/n$; unit: second
Loudness — Property distinguishing weak from loud sound; depends on amplitude; unit: decibel (dB)	Pitch — Property distinguishing flat from shrill sound; depends on frequency; higher freq = higher pitch
Quality/Timbre — Property that distinguishes two sounds of same pitch and amplitude from different sources	Audible Range — 20 Hz to 20,000 Hz — frequency range audible to normal human ears
Infrasonic Sound — Sound with frequency below 20 Hz — not audible to humans; detected by dogs, dolphins, elephants	Ultrasonic Sound — Sound with frequency above 20,000 Hz — not audible to humans; used in SONAR, medical imaging
Echo — Reflected sound heard distinctly after original sound; reflecting surface must be at least ~17 m away	Reverberation — Persistence of sound due to multiple reflections in closed spaces; sound continues after source stops

SONAR — Sound Navigation And Ranging — uses ultrasound to detect and measure distances underwater	Echolocation — Navigation by sending ultrasound pulses and detecting echoes — used by bats and dolphins
Wind Instrument — Musical instrument producing sound by vibrating air column in hollow tube — e.g., flute, trumpet	Reed Instrument — Musical instrument with reed that vibrates when air is blown — e.g., harmonium, mouth organ
Stringed Instrument — Musical instrument producing sound by vibrating string/wire — e.g., guitar, violin, sitar	Percussion Instrument — Musical instrument producing sound when struck, scraped or clashed — e.g., drum, tabla
Larynx — Voice box — organ in throat at upper end of windpipe; contains vocal cords	Vocal Cords — Two ligaments in larynx that vibrate when air passes through them to produce voice
Pinna — Outer visible curved part of ear that collects sound waves from environment	Eardrum (Tympanic Membrane) — Thin membrane that vibrates when sound waves strike it; between outer and middle ear
Ossicles — Three tiny bones in middle ear (Malleus, Incus, Stapes) that amplify and transmit vibrations	Cochlea — Spiral fluid-filled structure in inner ear that converts vibrations to nerve signals
Eustachian Tube — Tube connecting middle ear to throat — equalises air pressure on both sides of eardrum	Noise — Unwanted, irritating, unpleasant sound produced by irregular and non-periodic vibrations
Noise Pollution — Disturbance in environment due to loud, harsh sounds; bi-product of industrialisation	Phonograph — First device to record and play back sound — invented by Thomas Alva Edison in 1877
Seismology — Branch of science studying seismic waves (produced during earthquakes, explosions)	Galton's Whistle — Ultrasonic whistle inaudible to humans but heard by dogs — used in dog training
Sonogram — Medical imaging using ultrasound waves to visualise internal organs; non-invasive	

■ Did You Know?

- ◆ Bats use echolocation to navigate in total darkness — they produce ultrasonic cries (>20,000 Hz) and detect echoes to locate prey.
- ◆ Lightning travels at speed of light (3×10^8 m/s) while thunder travels at ~330 m/s in air — that is why lightning is seen before thunder is heard.
- ◆ On the Moon, two astronauts cannot talk directly because there is no atmosphere — sound needs a medium. They use radio waves.
- ◆ The speed of sound is about 880,000 times slower than the speed of light — that is why distant explosions are heard after they are seen.
- ◆ Whales communicate using infrasound — frequencies as low as 10–39 Hz — detectable over thousands of kilometres in ocean.